CPSC 340: Machine Learning and Data Mining

Nonlinear Regression

Last Time: Linear Regression

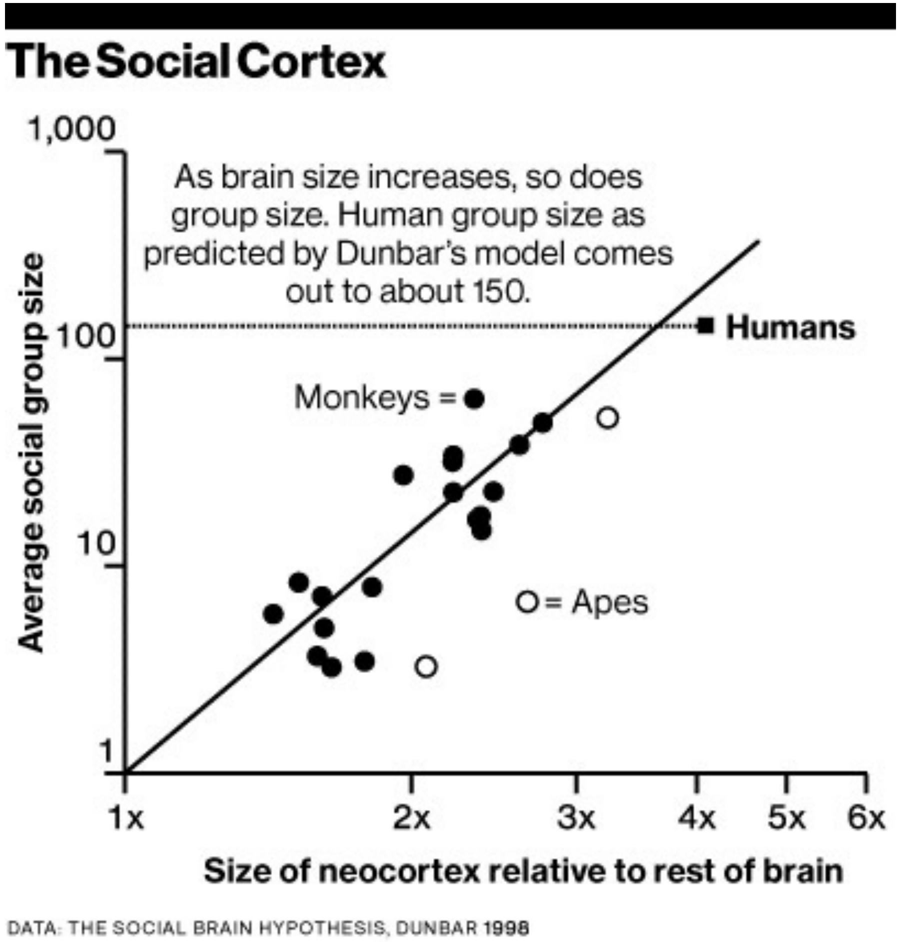
- We discussed linear models:

$$\hat{y}_i = w_1 x_{i1} + w_2 x_{i2} + \cdots + w_d x_{id}$$
$$= \sum_{j=1}^{d} w_j x_{ij}$$

- "Multiply feature x_{ij} by weight w_j, add them to get y_i".
- We discussed squared error function:

$$f(w) = \frac{1}{2} \sum_{i=1}^{n} (\underbrace{w^T x_i}_{\text{Predicted value}} - \underbrace{y_i}_{\text{True value}})^2$$

  - Minimize 'f' by equating gradient of 'f' with zero.
- Interactive demo:
  - http://setosa.io/ev/ordinary-least-squares-regression



To predict on test case $\tilde{x}_i$ use $\hat{y}_i = w^T \tilde{x}_i$

http://www.bloomberg.com/news/articles/2013-01-10/the-dunbar-number-from-the-guru-of-social-networks

Matrix/Norm Notation (MEMORIZE/STUDY THIS)

- Up to now: we have not been careful about row vs. column vectors (x_i was a “list”).
- From now on: we assume that vectors are column-vectors.
  - We use ‘w’ as a “d times 1” vector containing weight ‘w_j’ in position ‘j’.
  - We use ‘y’ as an “n times 1” vector containing target ‘y_i’ in position ‘i’.
  - We use ‘x_i’ as a “d times 1” vector containing features ‘j’ of example ‘i’.
    - We’re now going to be careful to make sure these are column vectors.
  - So ‘X’ is a matrix with x_i^T in row ‘i’.
- Recommended: see Course Notation Guide (on webpage).

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \qquad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \qquad x_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{id} \end{bmatrix} \qquad X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix} = \begin{bmatrix} \text{---} & x_1^T & \text{---} \\ \text{---} & x_2^T & \text{---} \\ & \vdots & \\ \text{---} & x_n^T & \text{---} \end{bmatrix}$$

Matrix/Norm Notation (MEMORIZE/STUDY THIS)

- We showed how to express various quantities in matrix notation:
  - Linear regression prediction for one example: $\hat{y}_i = w^T x_i$
  - Linear regression prediction for all 'n' examples: $\hat{y} = Xw$
  - Linear regression residual vector: $r = Xw - y$
  - Sum of residuals squared in linear regression model:

$$f(w) = \sum_{i=1}^{n} \left(\sum_{j=1}^{d} w_j x_{ij} - y_i \right)^2 = \|Xw - y\|^2$$

  - Today: derive gradient and least squares solution in matrix notation.

Digression: Matrix Algebra Review

- Quick review of linear algebra operations we'll use:
  - If 'a' and 'b' be vectors, and 'A' and 'B' be matrices then:

$$a^T b = b^T a$$

$$\|a\|^2 = a^T a$$

$$(A+B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

$$(A+B)(A+B) = AA + BA + AB + BB$$

$$a^T \underbrace{Ab}_{\text{vector}} = \underbrace{b^T A^T}_{\text{vector}} a$$

Sanity check:

ALWAYS CHECK THAT DIMENSIONS MATCH

(if not, you did something wrong)

Linear and Quadratic Gradients

- From these rules we have (see post-lecture slide for steps):

$$f(w) = \frac{1}{2}\sum_{i=1}^{n}(w^Tx_i - y_i)^2 = \frac{1}{2}\|Xw - y\|^2 = \frac{1}{2}w^T\underbrace{X^TX}_{\text{matrix 'A'}}w - w^T\underbrace{X^Ty}_{\text{vector 'b'}} + \underbrace{\frac{1}{2}y^Ty}_{\text{scalar 'c'}}$$

$$= \frac{1}{2}w^T\underbrace{A}_{X^TX}w + w^T\underbrace{b}_{-X^Ty} + \underbrace{c}_{\frac{1}{2}y^Ty}$$

These are scalars so dimensions match.

- How do we compute gradient?

Let's first do it with $d=1$:

$$f(w) = \frac{1}{2}waw + wb + c$$

$$= \frac{1}{2}aw^2 + wb + c$$

$$f'(w) = aw + b + 0$$

Here are the generalizations to 'd' dimensions:

$\nabla[c] = 0$ (zero vector)

$\nabla[w^Tb] = b$

$\nabla[\frac{1}{2}w^TAw] = Aw$ (if A is symmetric)

Full derivations are on webpage in notes on linear and quadratic gradients.

Linear and Quadratic Gradients

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$$f(w) = \frac{1}{2}\sum_{i=1}^{n}(w^Tx_i - y_i)^2 = \frac{1}{2}\|Xw - y\|^2 = \frac{1}{2}w^T\underbrace{X^TX}_{\text{matrix 'A'}}w - w^T\underbrace{X^Ty}_{\text{vector 'b'}} + \underbrace{\frac{1}{2}y^Ty}_{\text{scalar 'c'}}$$

$$= \frac{1}{2}w^T\underset{X^TX}{A}w + w^T\underset{-X^Ty}{b} + \underset{\frac{1}{2}y^Ty}{c}$$

- Gradient is given by: $\nabla f(w) = Aw + b + 0$

- Using definitions of 'A' and 'b': $= \underbrace{X^TXw - X^Ty}_{\substack{\text{Sanity check: all dimensions match} \\ (d\times n)(n\times d)(d\times 1) - (d\times n)(n\times 1)}}$

Normal Equations

- Set gradient equal to zero to find the "critical" points:

$$X^T X w - X^T y = 0$$

- We now move terms not involving 'w' to the other side:

$$X^T X w = X^T y$$

- This is a set of 'd' linear equations called the "normal equations".
  - This a linear system like "Ax = b" from Math 152 ('A' is $X^T X$, 'x' is 'w').
    - You can use Gaussian elimination to solve for 'w'.
  - In Python, you solve linear systems in 1 line using numpy.linalg.solve.

Incorrect Solutions to Least Squares Problem

The least squares objective is $f(w) = \frac{1}{2}\|Xw - y\|^2$

The minimizers of this objective are solutions to the linear system:

$$X^T X w = X^T y$$

The following are not the solutions to the least squares problem:

$w = (X^T X)^{-1}(X^T y)$ (only true if $X^T X$ is invertible)

$w X^T X = X^T y$ (matrix multiplication is not commutative, dimensions don't even match)

$w = \frac{X^T y}{X^T X}$ (you cannot divide by a matrix)

Least Squares Cost

- Cost of solving "normal equations" $X^TXw = X^Ty$?
- Forming X^Ty vector costs $O(nd)$.
  - It has 'd' elements, and each is an inner product between 'n' numbers.
- Forming matrix X^TX costs $O(nd^2)$.
  - It has d^2 elements, and each is an inner product between 'n' numbers.
- Solving a d x d system of equations costs $O(d^3)$.
  - Cost of Gaussian elimination on a d-variable linear system.
  - Other standard methods have the same cost.
- Overall cost is $O(nd^2 + d^3)$.
  - Which term dominates depends on 'n' and 'd'.

Least Squares Issues

- Issues with least squares model:
  - Solution might not be unique.
  - It is sensitive to outliers.
  - It always uses all features.
  - Data might so big we cannot store X^TX.
    - If you have 10 million features, this requires $O(d^2)$.
    - Or you cannot afford the $O(nd^2 + d^3)$ cost.
  - It might predict outside range of y_i values.
    - For some applications, only positive y_i values are valid.
  - It assumes a linear relationship between x_i and y_i.

X is $n \times d$

so X^T is $d \times n$

and X^TX is $d \times d$.

Non-Uniqueness of Least Squares Solution

- Why is the solution vector 'w' not unique?
  - Imagine having two features that are identical for all examples.
  - I can increase weight on one feature, and decrease it on the other, without changing predictions.

$$\hat{y}_i = w_1 x_{i1} + w_2 \underbrace{x_{i1}}_{\text{copy}} = (w_1 + w_2) x_{i1} + 0 x_{i1}$$

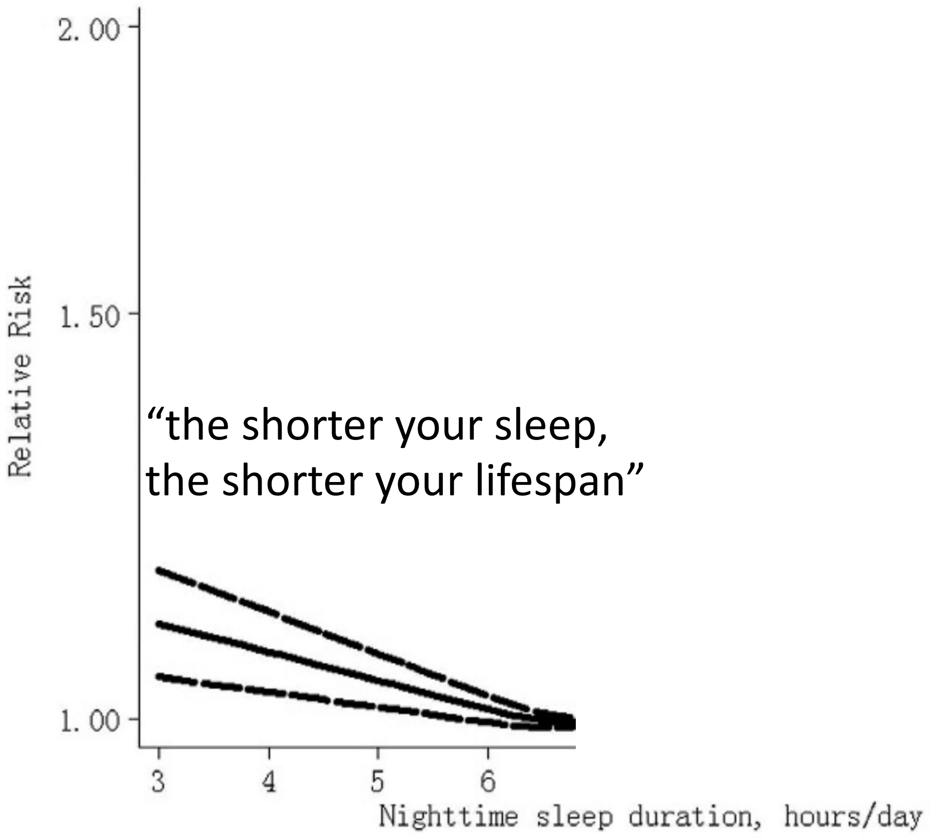
  - In this setting, if (w_1, w_2) is a solution then $(w_1 + w_2, 0)$ is another solution.
  - This is special case of features being "collinear":
    - One feature is a linear function of the others.

- But, any 'w' where $\nabla f(w) = 0$ is a global minimizer of 'f'.
  - This is due to convexity of 'f', which we will discuss later.

Next Topic: Non-Linear Regression

Motivation: Non-Linear Regression

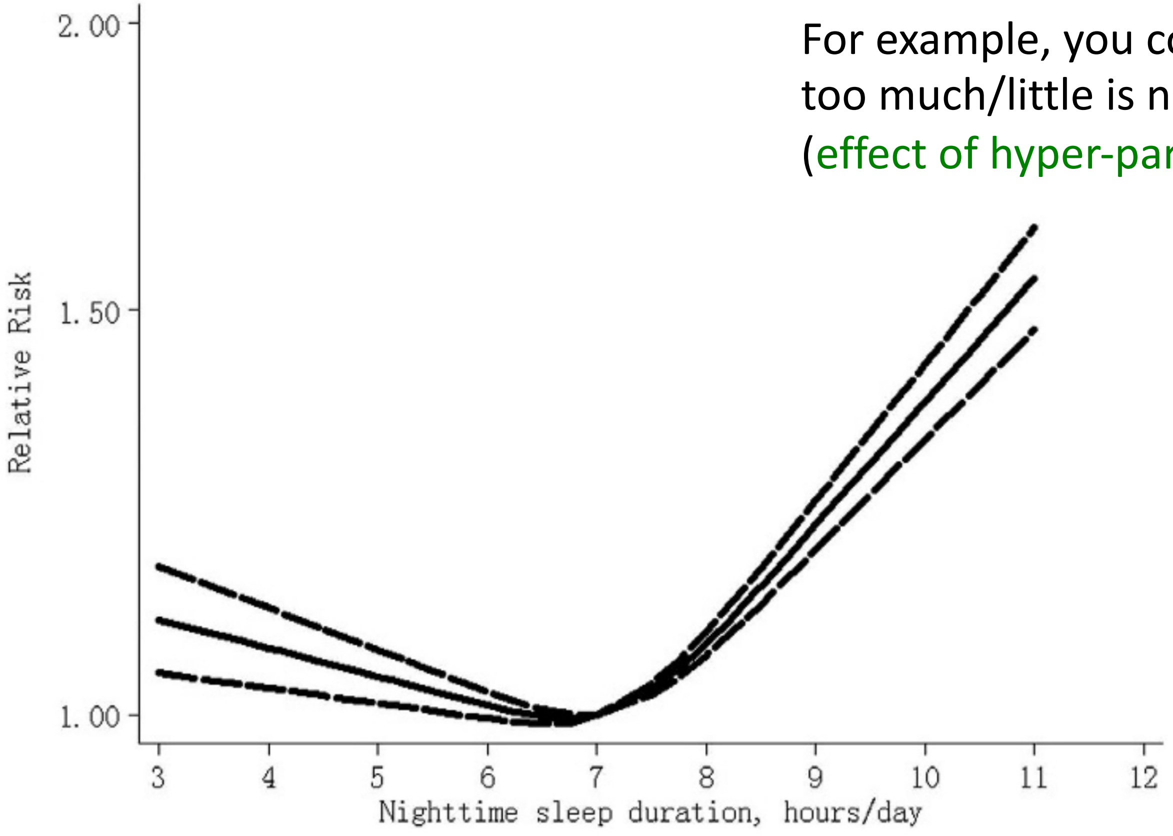
- Many relationships are approximated well by linear function.



https://www.nature.com/articles/srep21480.pdf

Motivation: Non-Linear Regression

- Many relationships are approximated well by linear function.
  - But many are also highly non-linear.



For example, you could have a "u-shape" when too much/little is not good.
(effect of hyper-parameters usually looks like this)

https://www.nature.com/articles/srep21480.pdf

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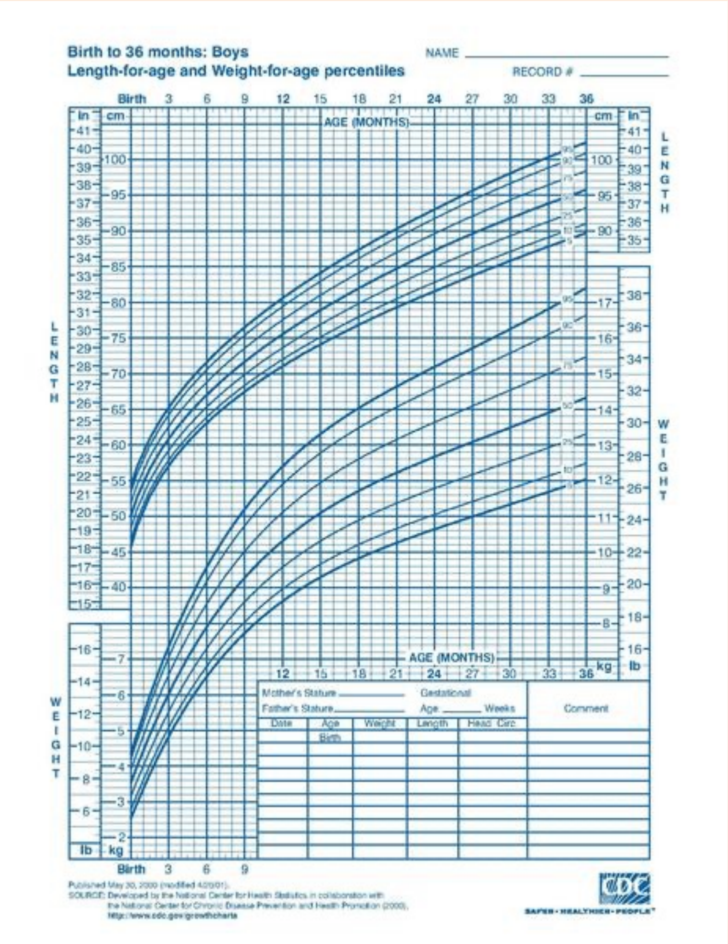

Slope could slowly change
or reach asymptote.

https://commons.wikimedia.org/wiki/File:CDC_growth_chart_b

Motivation: Non-Linear Regression

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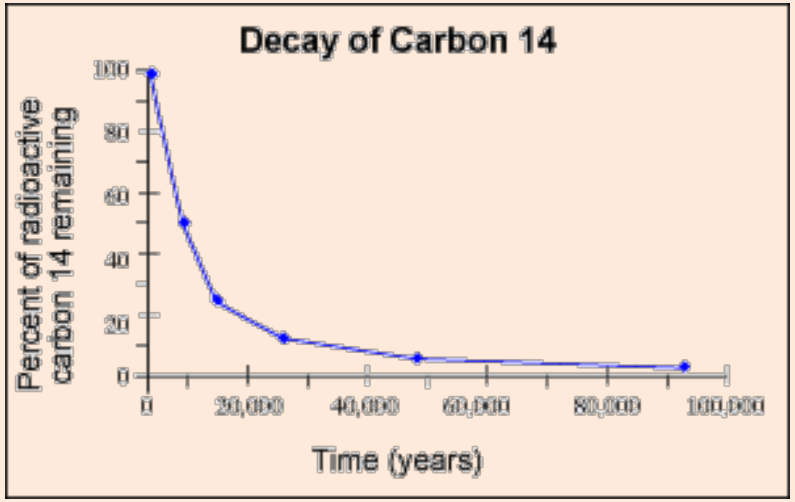
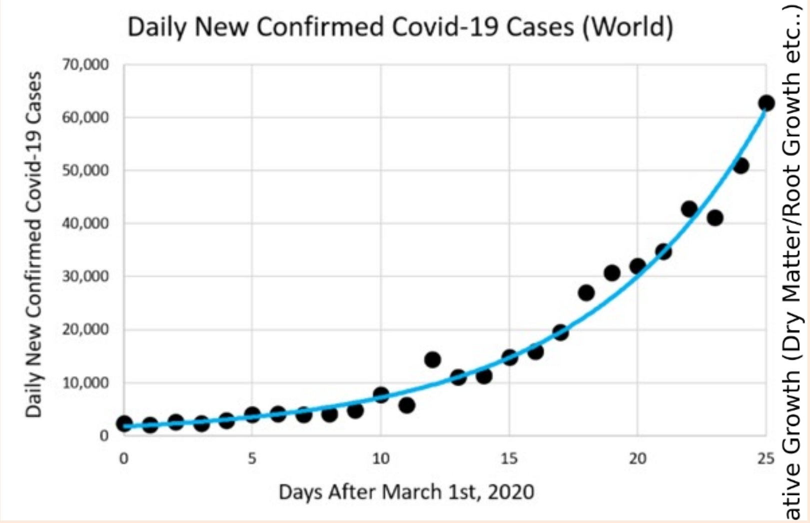
  - But many are also highly non-linear.

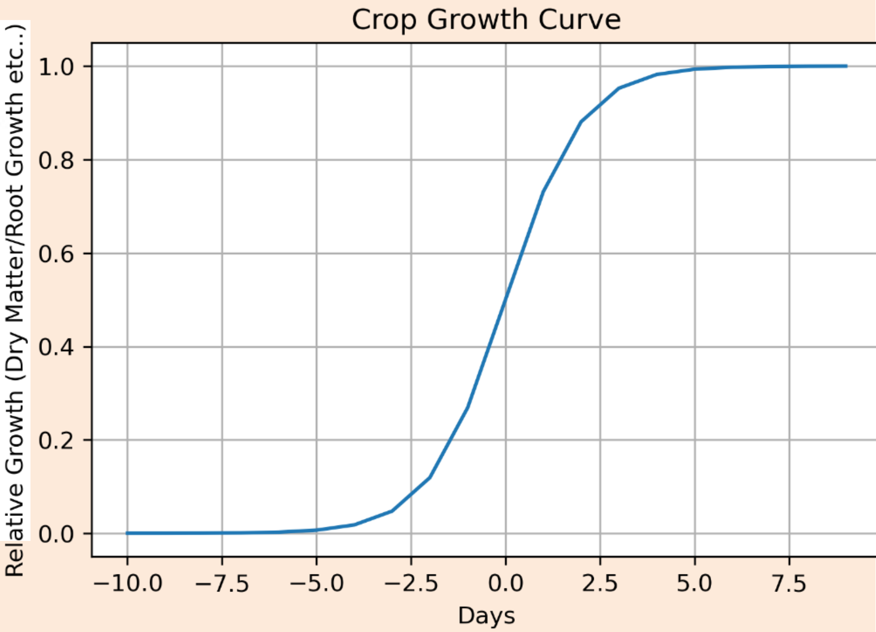
"geometric decay"



"exponential growth"



"sigmoidal growth".

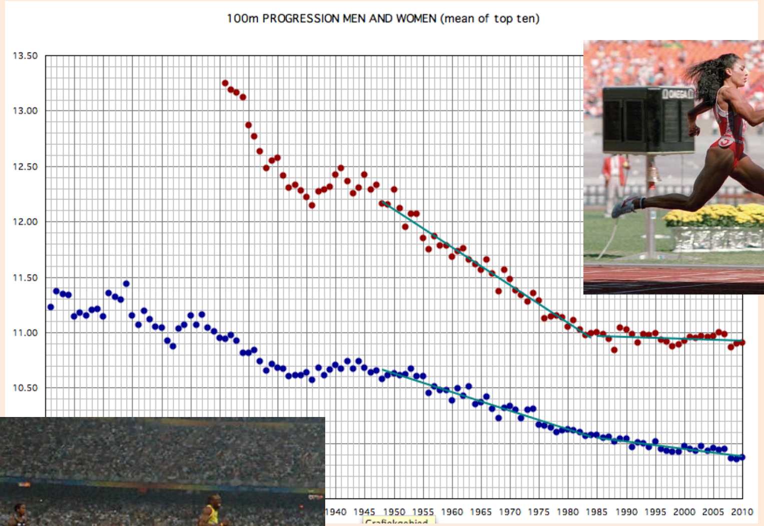


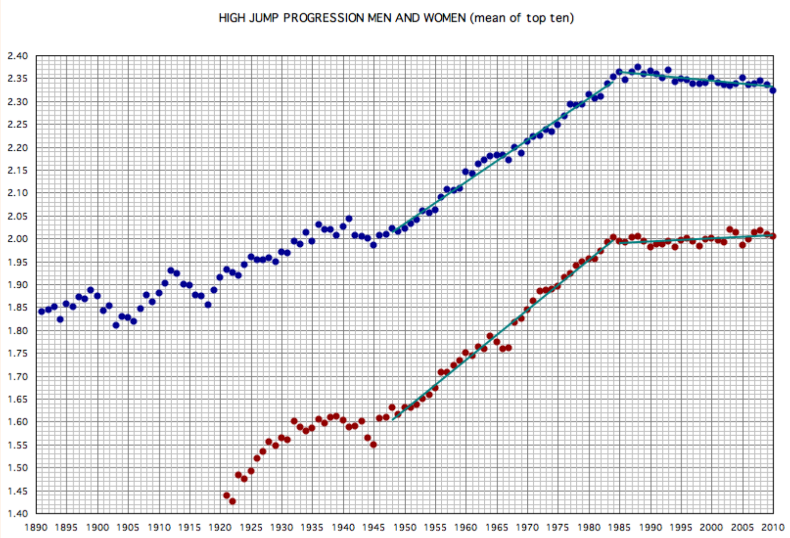
http://www.biology.arizona.edu/biomath/tutorials/Applications/Carbon.html
https://www.nytimes.com/2021/05/14/learning/the-math-of-ending-the-pandemic-exponential-growth-and-decay.html
https://stackoverflow.com/questions/69535905/how-to-dynamically-fit-sigmoid-growth-curve-for-crop-plants

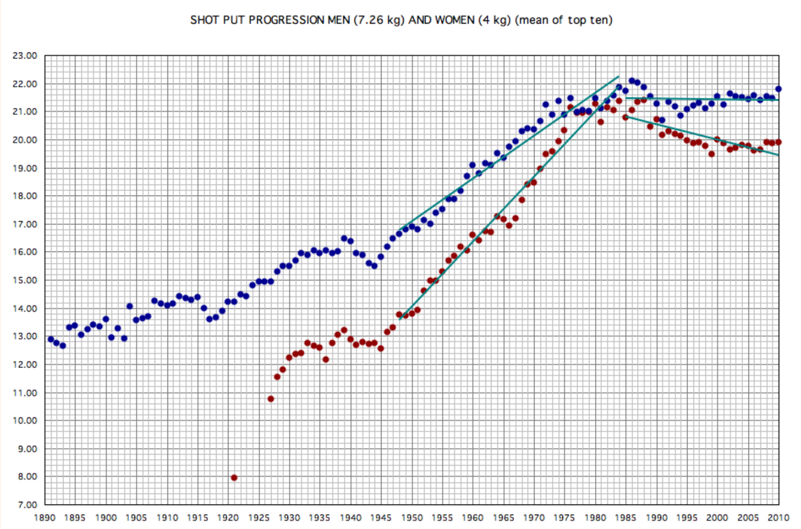
Motivation: Non-Linear Regression

- Many relationships are approximated well by linear function.
  - But many are also highly non-linear.

"Piecewise linear": different pieces follow different linear functions.
(or be linear up to asymptote or phase transition)







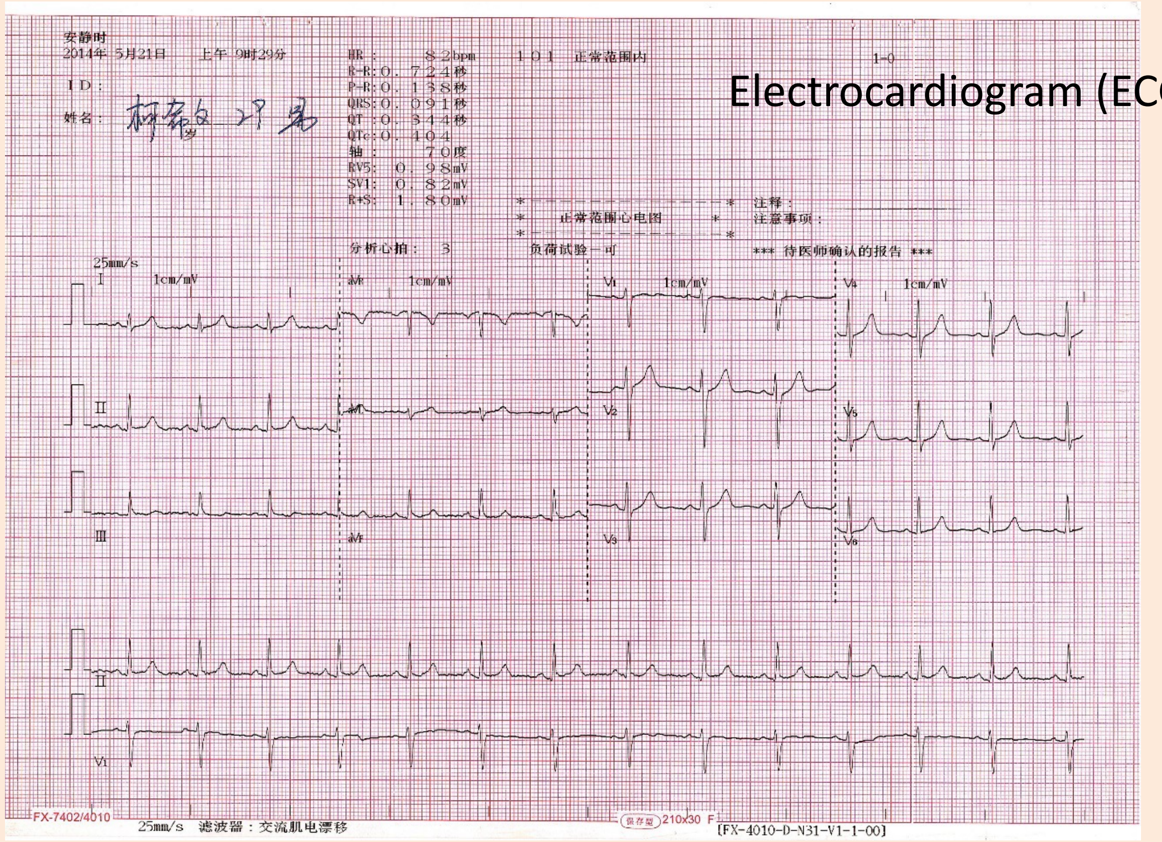
100 meter times will not go negative!

http://www.at-a-lanta.nl/weia/Progressie.html
https://en.wikipedia.org/wiki/Usain_Bolt
http://www.britannica.com/biography/Florence-Griffith-Joyner

Motivation: Non-Linear Regression

- Many relationships are approximated well by linear function.
  - But many are also highly non-linear.

"Periodic" signals.



Electrocardiogram (ECG)

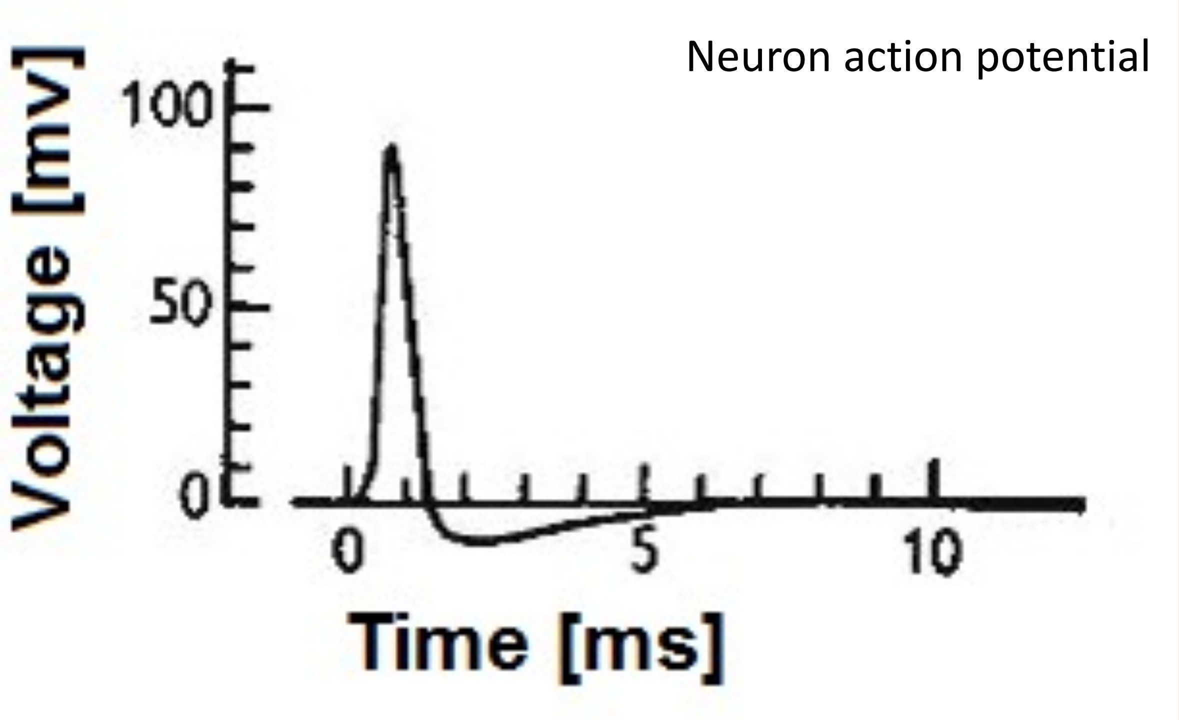
Electroencephalography (EEG)

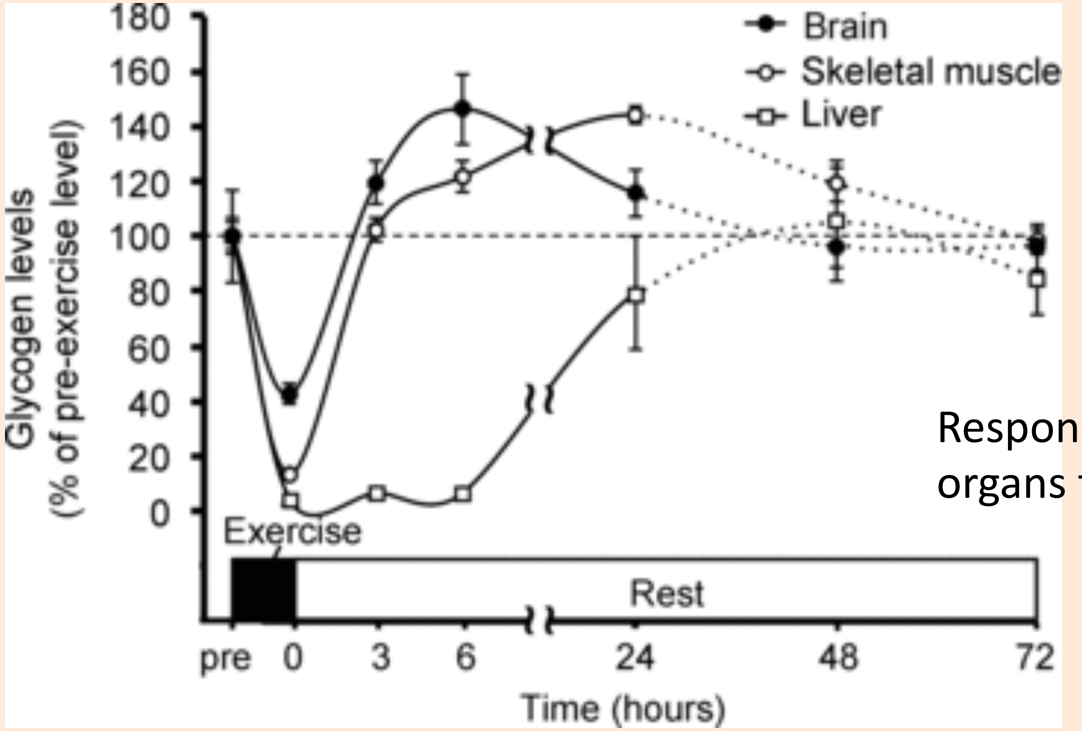
https://en.wikipedia.org/wiki/Electrocardiography#/media/File:Normal_12_lead_EKG.jpg

Motivation: Non-Linear Regression

- Many relationships are approximated well by linear function.
  - But many are also highly non-linear.

"Spike then recover"





Response of organs to exercise

https://en.wikipedia.org/wiki/Biological_neuron_model
https://physoc.onlinelibrary.wiley.com/doi/full/10.1113/jphysiol.2011.217919

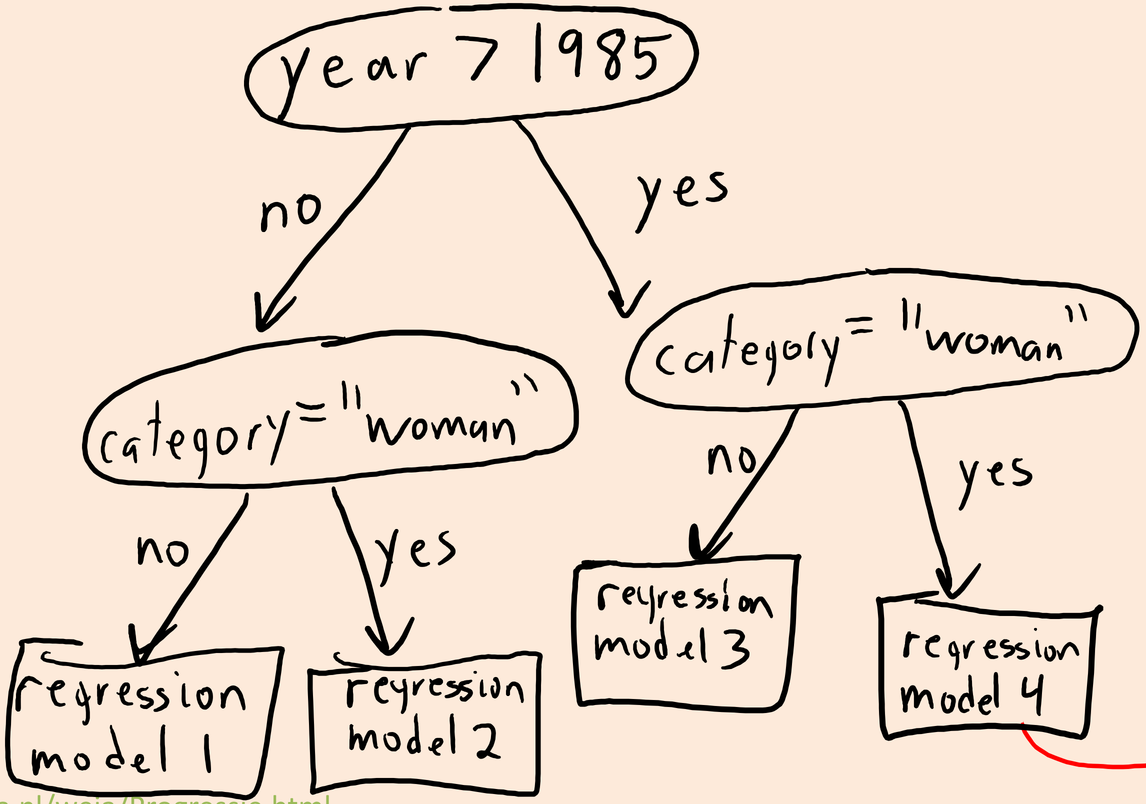
Adapting Counting/Distance-Based Methods

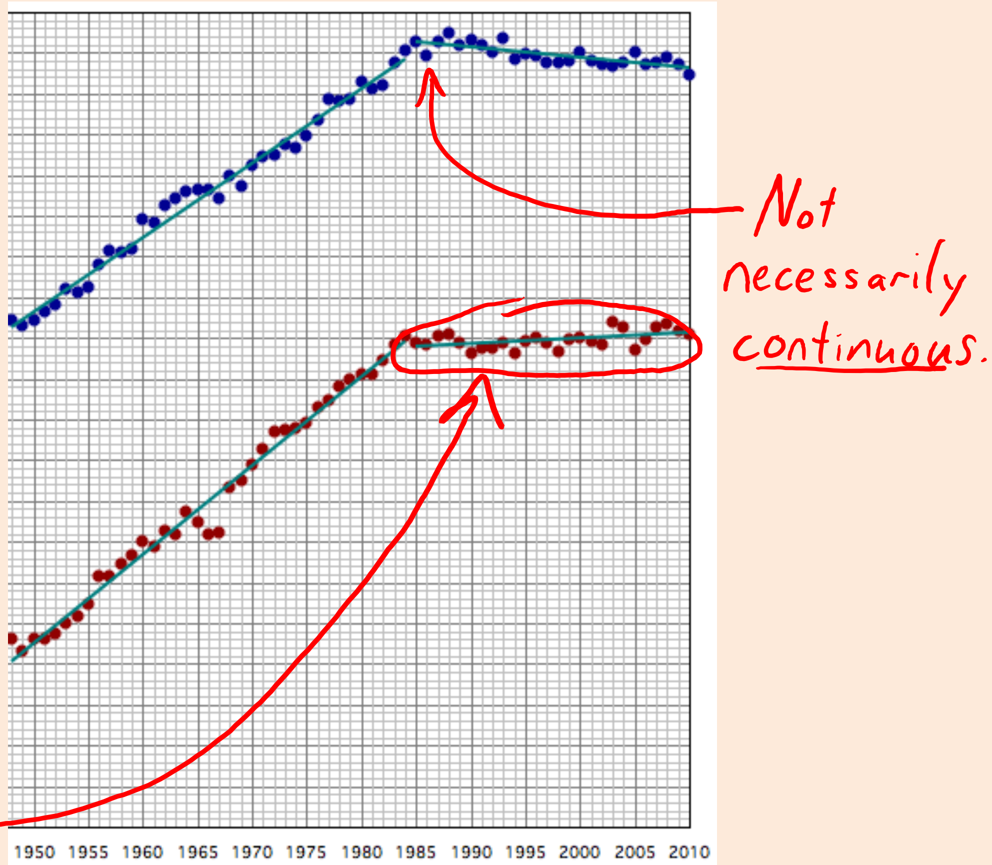
- Can adapt classification methods to perform non-linear regression:

http://www.at-a-lanta.nl/weia/Progressie.html

Adapting Counting/Distance-Based Methods

- Can adapt classification methods to perform non-linear regression:
  - Regression tree: tree with mean value or linear regression at leaves.





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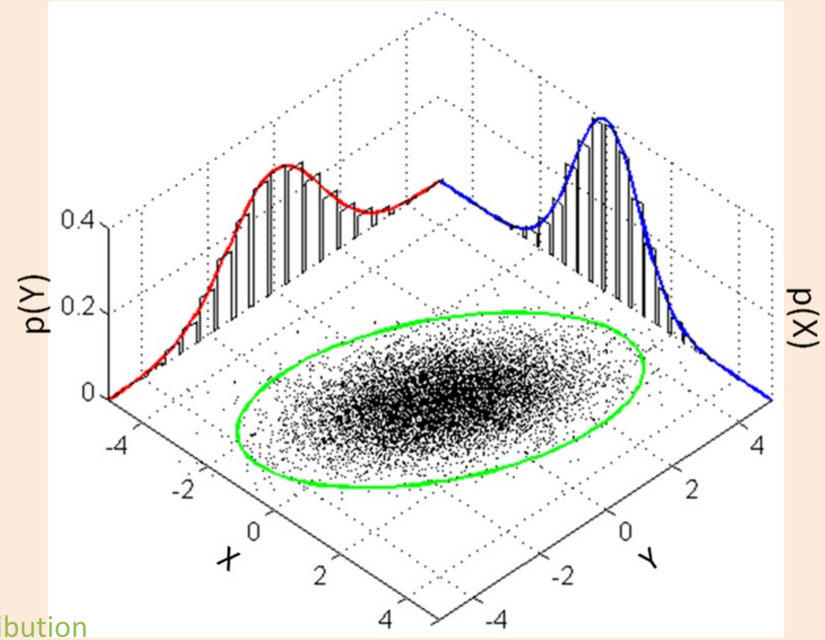
Adapting Counting/Distance-Based Methods

- Can adapt classification methods to perform non-linear regression:
  - Regression tree: tree with mean value or linear regression at leaves.
  - Probabilistic models: fit $p(x_i \mid y_i)$ and $p(y_i)$ with Gaussian or other model.
    - Take CPSC 440.



https://en.wikipedia.org/wiki/Multivariate_normal_distribution

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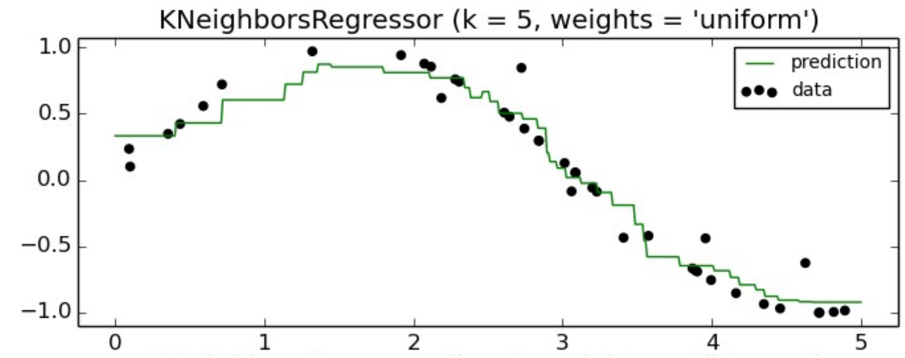
  - Probabilistic models: fit $p(x_i \mid y_i)$ and $p(y_i)$ with Gaussian or other model.
  - Non-parametric models:
    - KNN regression:
      - Find 'k' nearest neighbours of $\tilde{x}_i$.
      - Return the mean of the corresponding y_i.



http://scikit-learn.org/stable/modules/neighbors.html

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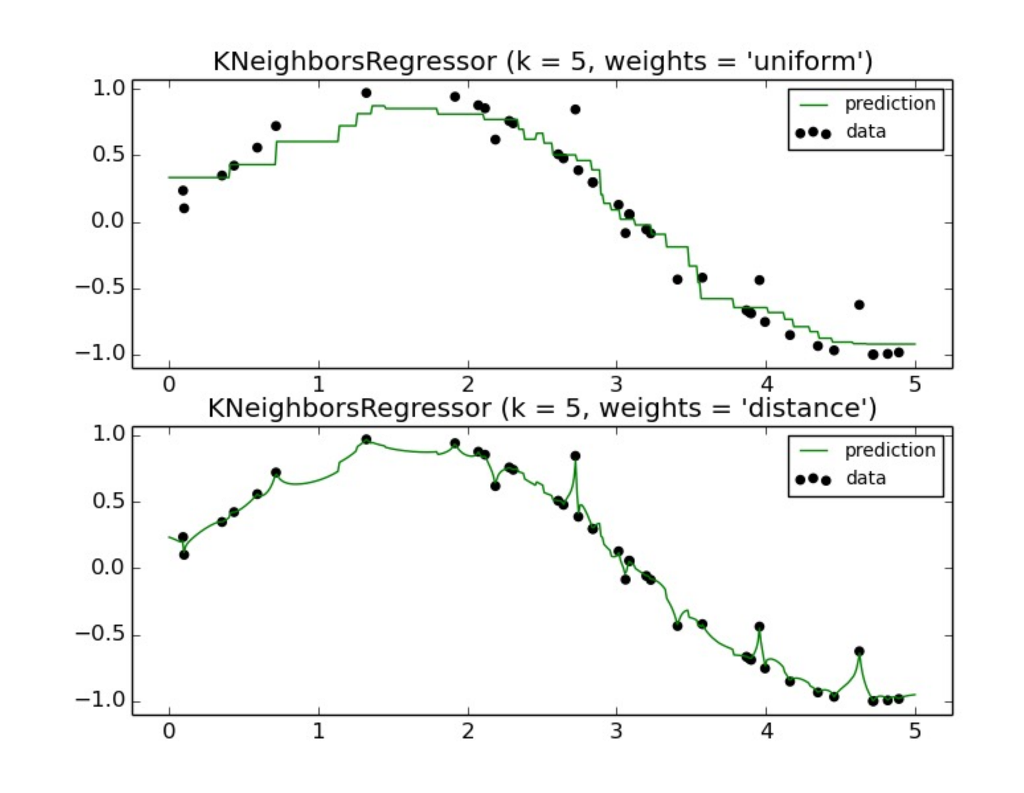
  - Non-parametric models:
    - KNN regression.
    - Could be weighted by distance.
      - Close points 'j' get more "weight" w_{ij}.



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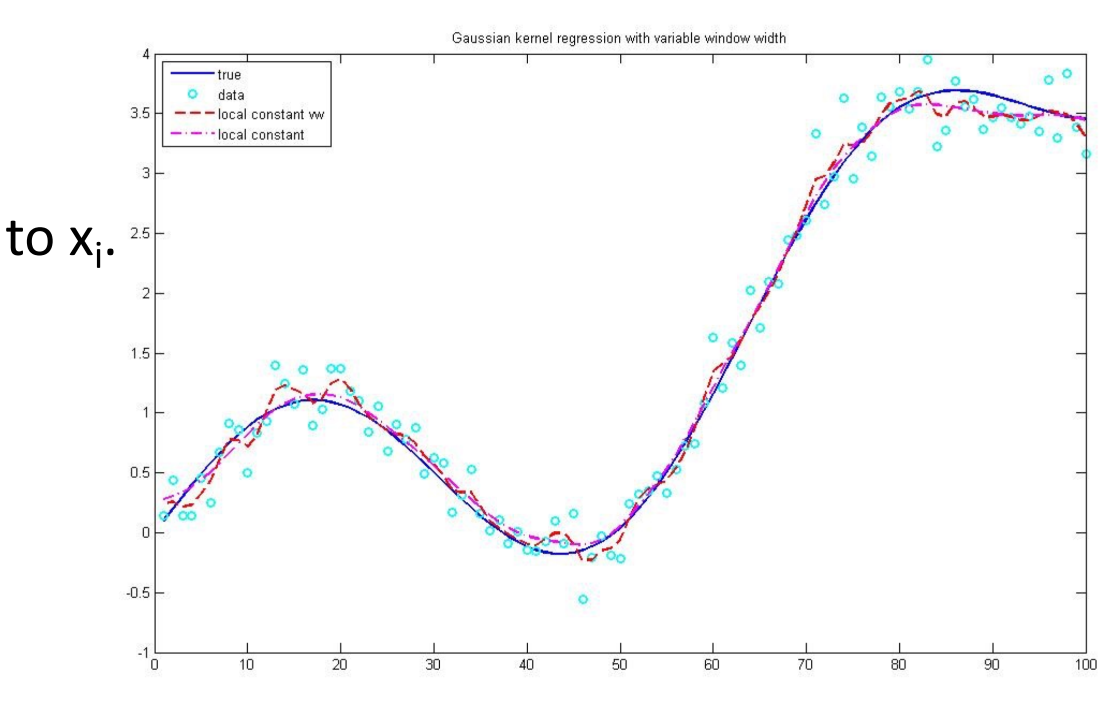
  - Non-parametric models:
    - KNN regression.
    - Could be weighted by distance.
    - 'Nadaraya-Watson': weight *all* y_i by distance to x_i.

$$\hat{y}_i = \frac{\sum_{j=1}^{n} v_{ij} y_j}{\sum_{j=1}^{n} v_{ij}}$$



http://www.mathworks.com/matlabcentral/fileexchange/35316-kernel-regression-with-variable-window-width/content/ksr_vw.m

Adapting Counting/

- Can adapt classification meth
  - Regression tree: tree with mea
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    - KNN regression.
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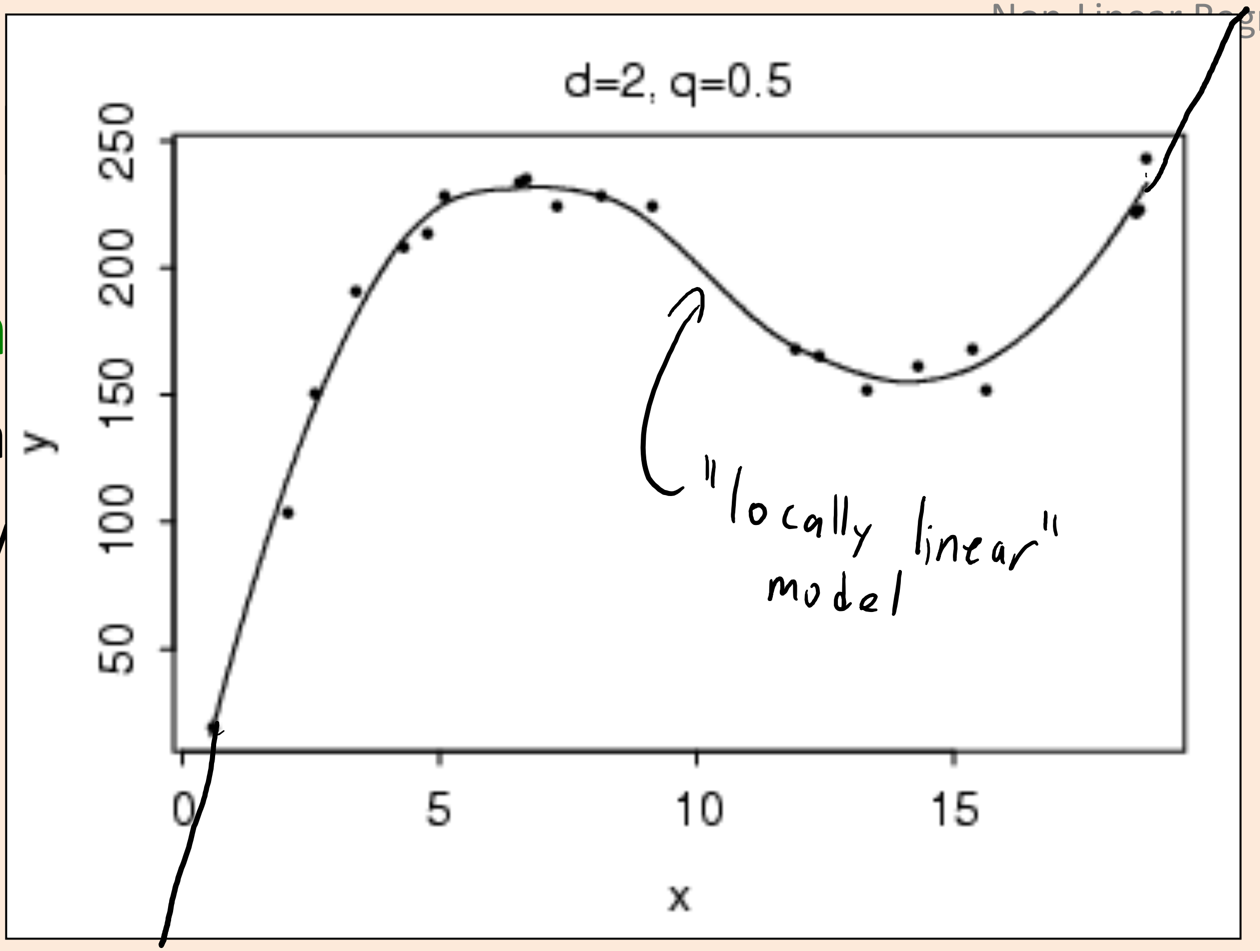
    - 'Nadaraya-Watson': weight *all* y_i
    - '**Locally linear regression**': for each x_i, fit a linear model weighted by distance.

(Better than KNN and NW at boundaries.)



http://www.itl.nist.gov/div898/handbook/pmd/section4/pmd423.htm

Adapting Counting/Distance-Based Methods

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    - 'Nadaraya-Watson': weight *all* y_i by distance to x_i.
    - 'Locally linear regression': for each x_i, fit a linear model weighted by distance.
      (Better than KNN and NW at boundaries.)
  - Ensemble methods:
    - Can improve performance by averaging predictions across regression models.

Adapting Counting/Distance-Based Methods

- Applications of non-linear regression (we will see many more):
  - Regression forests for fluid simulation:
  - KNN for image completion:
    - Combined with “graph cuts” and “Poisson blending”.
    - See also “PatchMatch”.
  - KNN regression for “voice photoshop”:
    - Combined with “dynamic time warping” and “Poisson blending”.

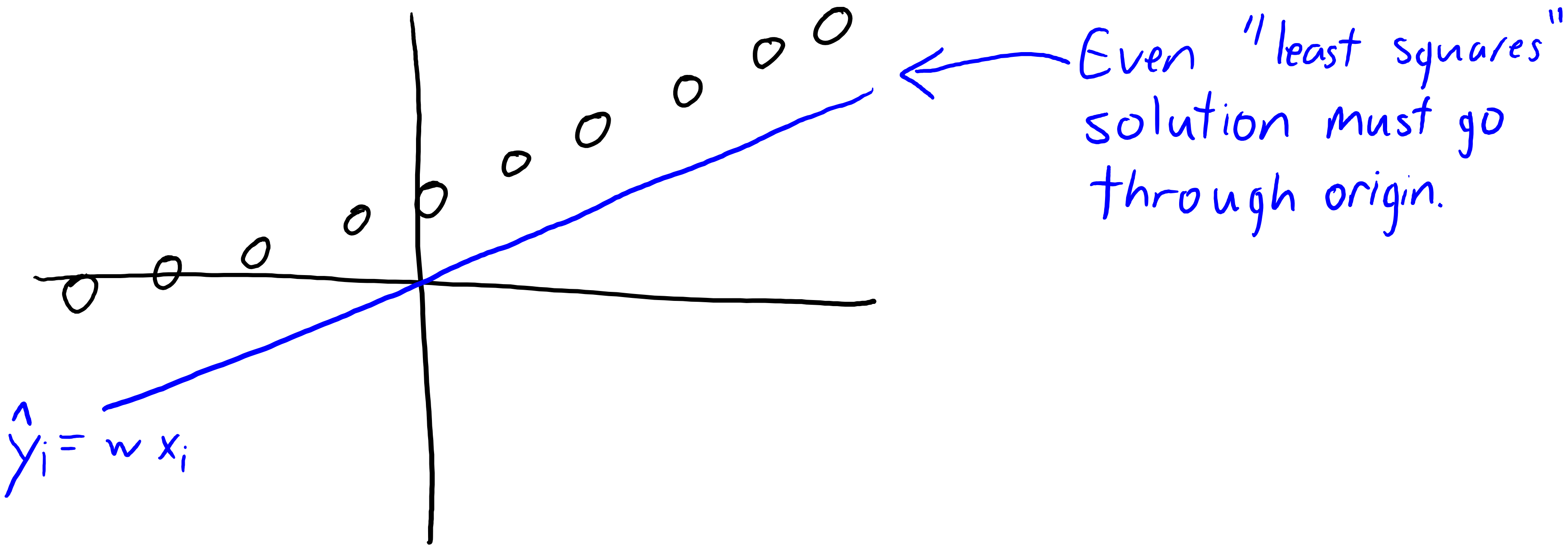
- We will first focus on linear models with non-linear transforms.
  - These are the building blocks for more advanced methods.

http://www.itl.nist.gov/div898/handbook/pmd/section4/pmd423.htm

Why don't we have a y-intercept?

– Linear model is $\hat{y}_i = wx_i$ instead of $\hat{y}_i = wx_i + w_0$ with y-intercept w_0.

– Without an intercept, if $x_i = 0$ then we must predict $\hat{y}_i = 0$.



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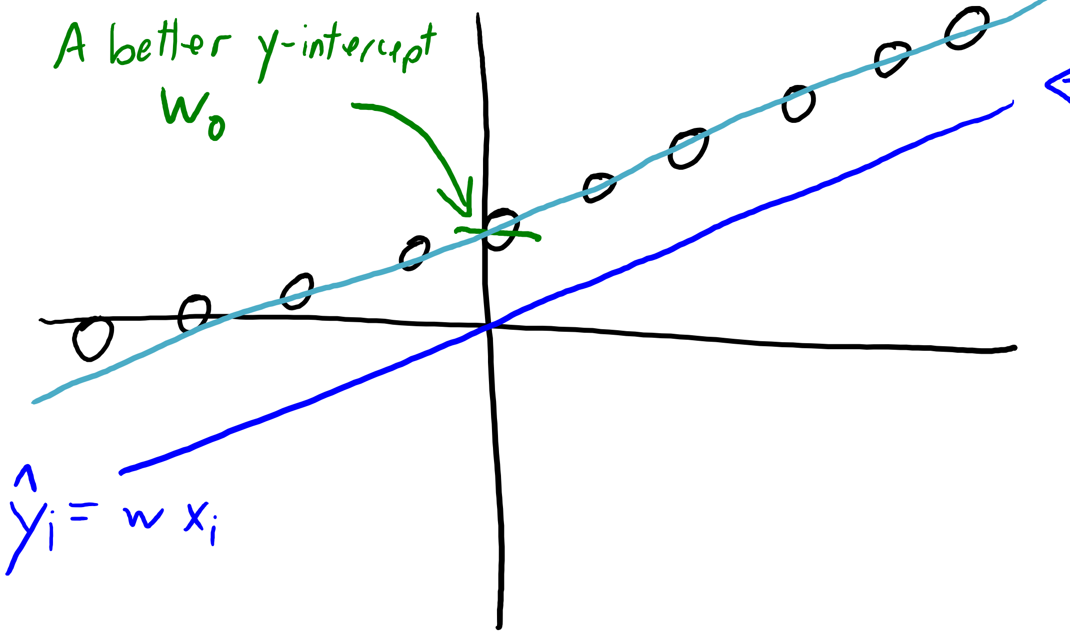
- Without an intercept, if $x_i = 0$ then we must predict $\hat{y}_i = 0$.

Adding y-intercept fixes this.

$\hat{y}_i = w^T x_i + w_0$



Even "least squares" solution must go through origin.

Adding a Y-Intercept ("Bias") Variable

- Simple trick to add a y-intercept ("bias") variable:
  - Make a new matrix "Z" with an extra feature that is always "1".

$$X = \begin{bmatrix} -0.1 \\ 0.3 \\ 0.2 \end{bmatrix} \qquad Z = \begin{bmatrix} \underbrace{\begin{matrix} 1 \\ 1 \\ 1 \end{matrix}}_{\text{"always 1"}} & \underbrace{\begin{matrix} -0.1 \\ 0.3 \\ 0.2 \end{matrix}}_{X} \end{bmatrix}$$

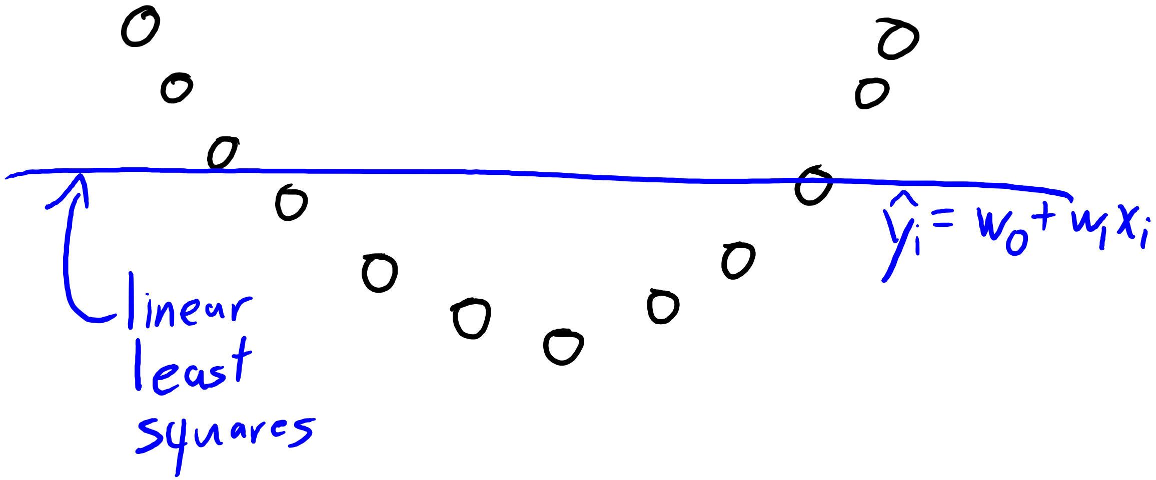
- Now use "Z" as your features in linear regression.
  - We will use 'v' instead of 'w' as regression weights when we use features 'Z'.

$$\hat{y}_i = \underset{\downarrow\, w_0}{v_1}\, \underset{\downarrow\, 1}{z_{i1}} + \underset{\downarrow\, w_1}{v_2}\, \underset{\downarrow\, x_{i1}}{z_{i2}} = w_0 + w_1 x_{i1}$$

- So we can have a non-zero y-intercept by changing features.
  - This means we can ignore the y-intercept to make cleaner derivations/code.

Motivation: Limitations of Linear Models

- On many datasets, y_i is not a linear function of x_i.



- A quadratic function would be a better fit for this dataset.

Non-Linear Feature Transforms

- Can we use linear least squares to fit a quadratic model?

$$\hat{y}_i = w_0 + w_1 x_i + w_2 x_i^2$$

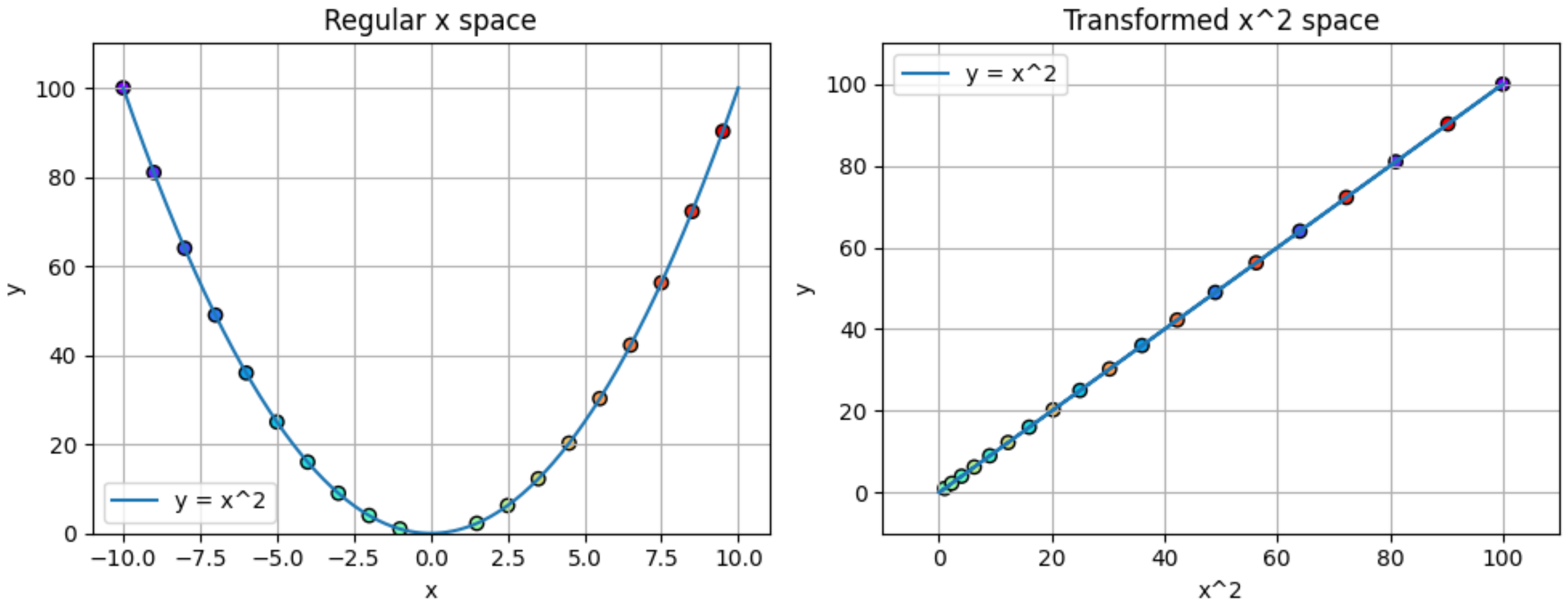
  - Notice that this is a non-linear function of x_i but a linear function of 'w'.
- So you can implement this by changing the features:

$$X = \begin{bmatrix} 0.2 \\ -0.5 \\ 1 \\ 4 \end{bmatrix} \qquad Z = \begin{array}{c} \begin{bmatrix} 1 & 0.2 & (0.2)^2 \\ 1 & -0.5 & (-0.5)^2 \\ 1 & 1 & (1)^2 \\ 1 & 4 & (4)^2 \end{bmatrix} \\ \begin{array}{ccc} \text{y-int} & x & x^2 \end{array} \end{array}$$

  - Fit new parameters 'v' under "change of basis": solve $Z^TZv = Z^Ty$.
- It's a linear function of w, but a quadratic function of x_i.

$$\hat{y}_i = v^T z_i = \underbrace{v_1}_{w_0} \underbrace{z_{i1}}_{1} + \underbrace{v_2}_{w_1} \underbrace{z_{i2}}_{x_i} + \underbrace{v_3}_{w_2} \underbrace{z_{i3}}_{x_i^2}$$

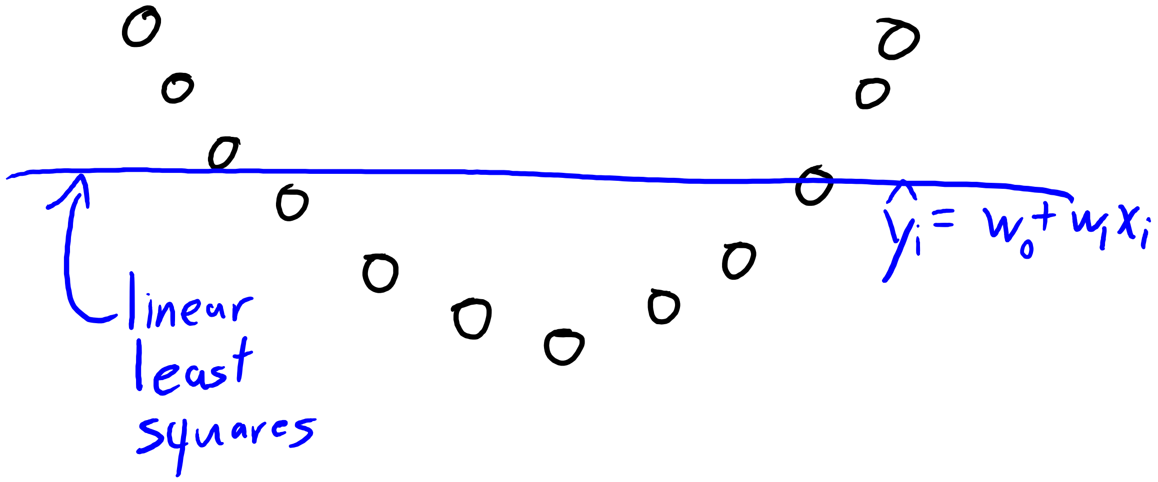
Non-Linear Feature Transforms

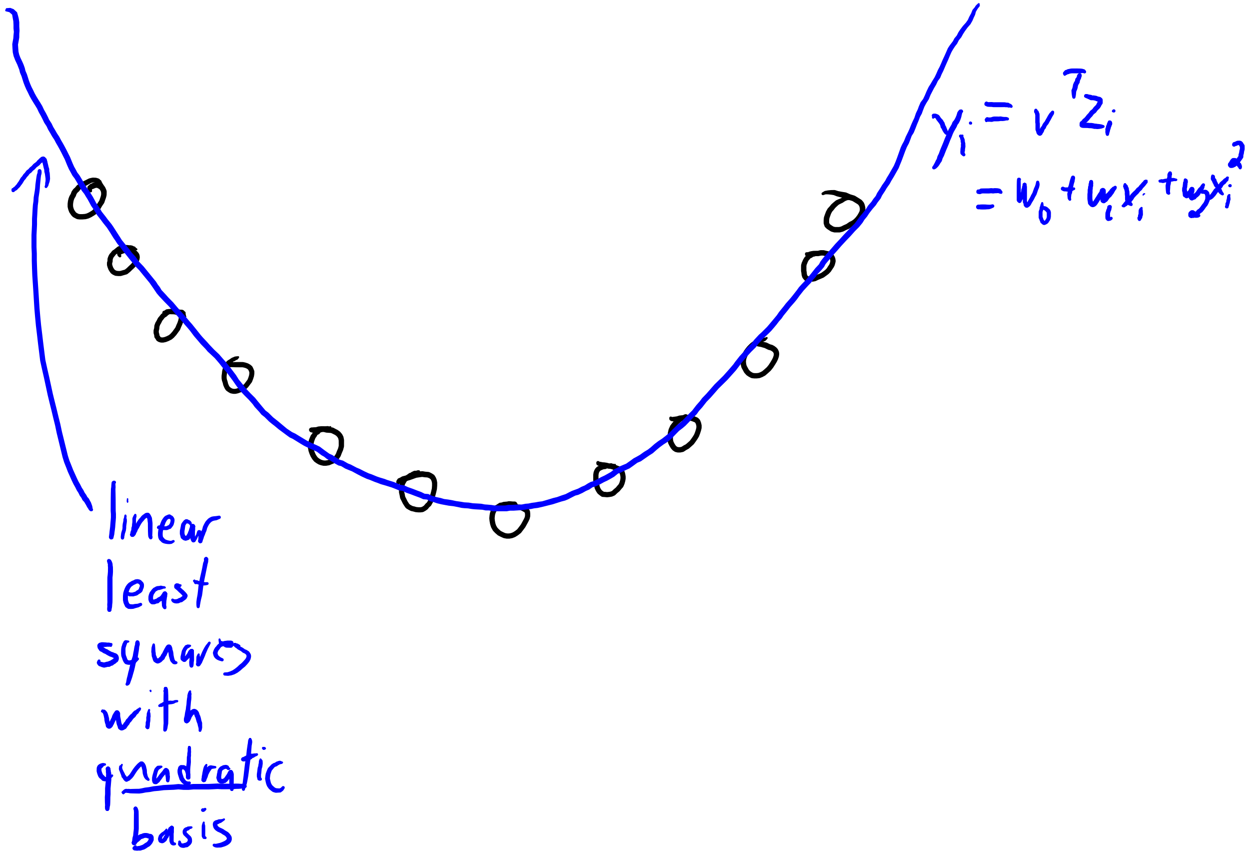


- It's a linear function of w, but a quadratic function of x_i.

$$\hat{y}_i = v^T z_i = \underbrace{v_1}_{w_0}\underbrace{z_{i1}}_{1} + \underbrace{v_2}_{w_1}\underbrace{z_{i2}}_{x_i} + \underbrace{v_3}_{w_2}\underbrace{z_{i3}}_{x_i^2}$$

Non-Linear Feature Transforms





To predict on new data $\tilde{X}$, form $\tilde{Z}$ from $\tilde{X}$ and take $y = \tilde{Z}v$

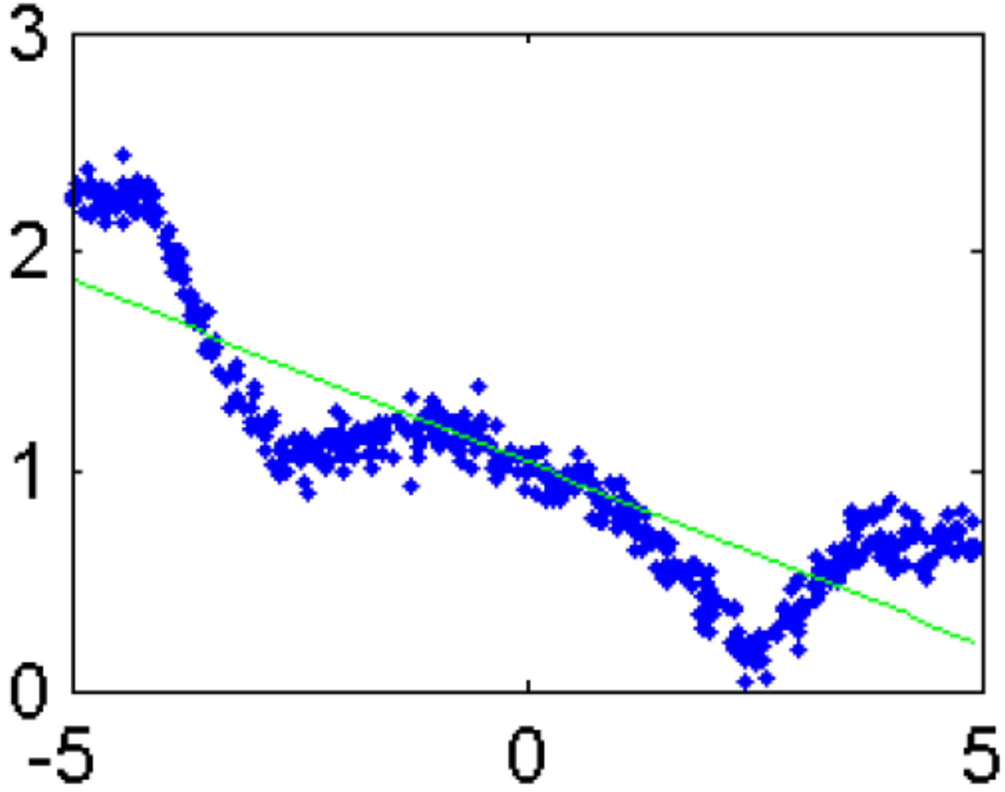
General Polynomial Features (d=1)

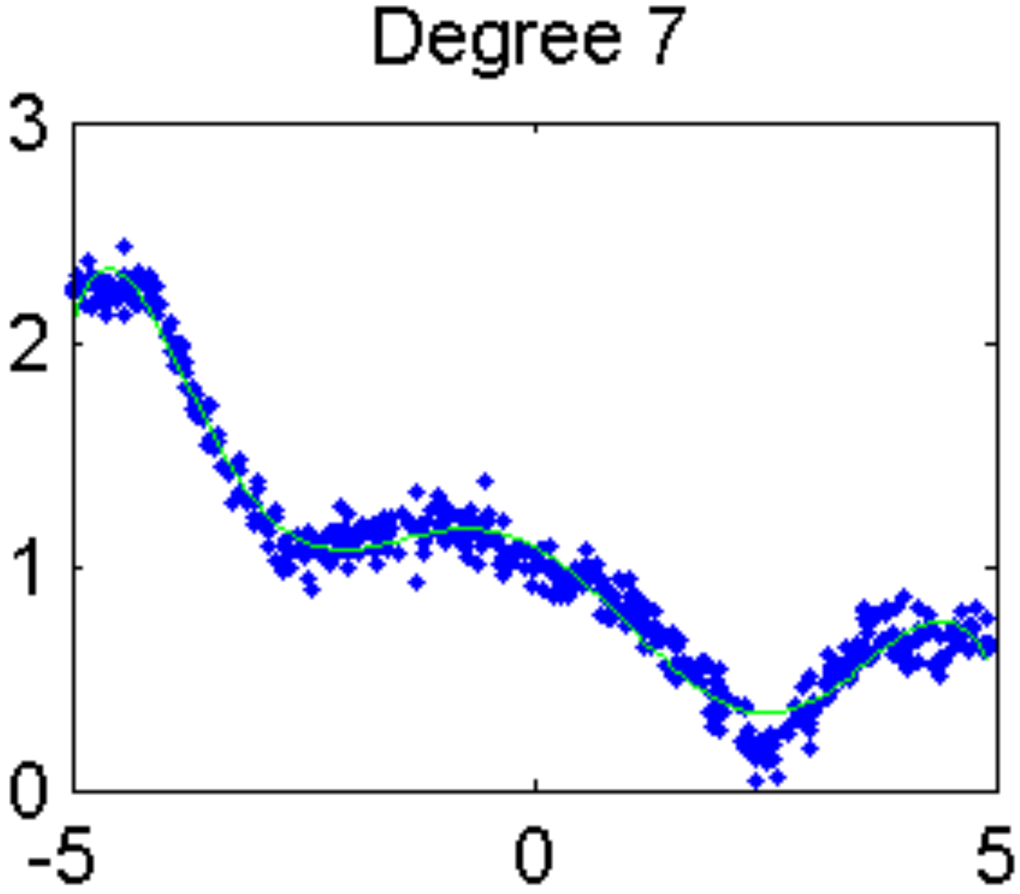
- We can have a polynomial of degree 'p' by using these features:

$$Z = \begin{bmatrix} 1 & x_1 & (x_1)^2 & \cdots & (x_1)^p \\ 1 & x_2 & (x_2)^2 & \cdots & (x_2)^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & (x_n)^2 & \cdots & (x_n)^p \end{bmatrix}$$

- There are polynomial basis functions that are numerically nicer:
  - Such as Lagrange polynomials (see CPSC 303).

General Polynomial Features





- If you have more than one feature, you can include interactions:
  - With p=2, in addition to $(x_{i1})^2$ and $(x_{i2})^2$ you could include $x_{i1}x_{i2}$.

"Change of Basis" Terminology

- Instead of "nonlinear feature transform", in machine learning it is common to use the expression "change of basis".
  - The z_i are the "coordinates in the new basis" of the training example.

- "Change of basis" means something different in math:
  - Math: basis vectors must be linearly independent (in ML we don't care).
  - Math: change of basis must span the same space (in ML we change space).

- Unfortunately, saying "change of basis" in ML is common.
  - When I say "change of basis", just think "nonlinear feature transform".

Linear Basis vs. Nonlinear Basis

Usual linear Regression

Train:

- Use 'X' and 'y' to find 'w'

Test:

- Use $\tilde{X}$ and 'w' to find $\hat{y}$

Linear regression with change of basis

Train:

- Use 'X' to find 'Z'
- Use 'Z' and 'y' to find 'v'

Test:

- Use '$\tilde{X}$' to find '$\tilde{Z}$'
- Use $\tilde{Z}$ and 'v' to find $\hat{y}$

Change of Basis Notation (MEMORIZE)

- Linear regression with original features:
  - We use 'X' as our "n by d" data matrix, and 'w' as our parameters.
  - We can find d-dimensional 'w' by minimizing the squared error:

$$f(w) = \frac{1}{2}\|Xw - y\|^2$$

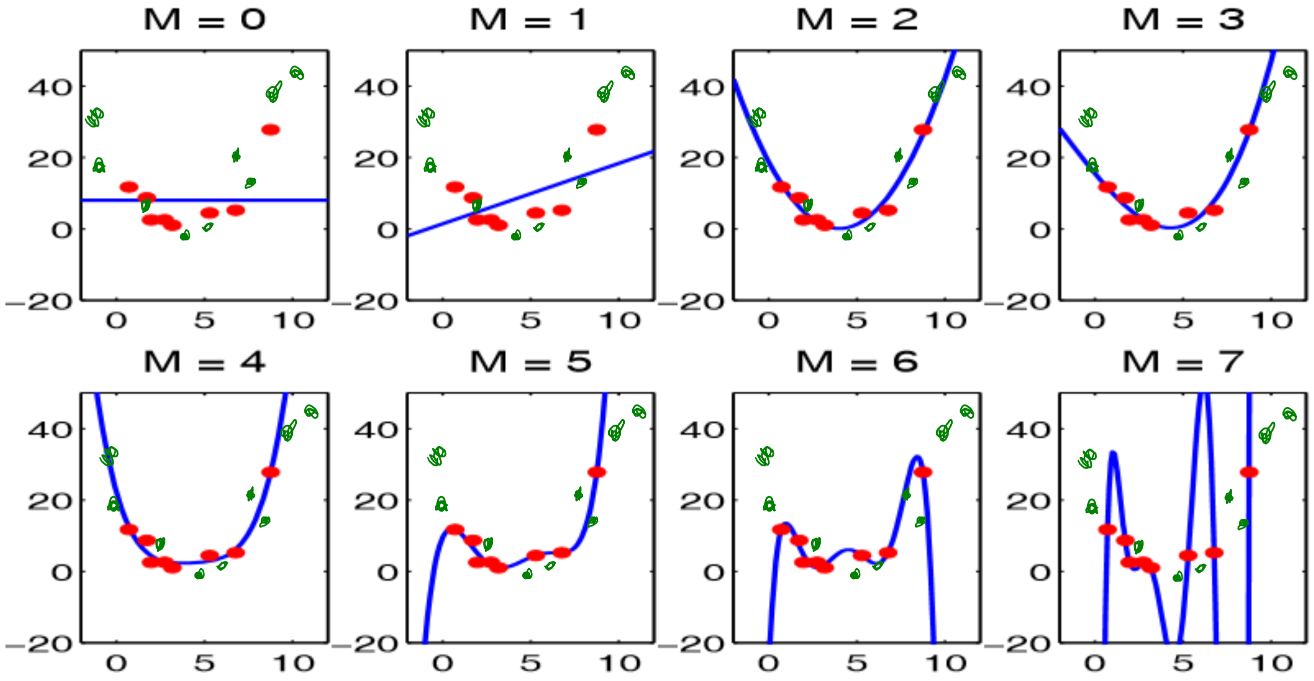
- Linear regression with nonlinear feature transforms:
  - We use 'Z' as our "n by k" data matrix, and 'v' as our parameters.
  - We can find k-dimensional 'v' by minimizing the squared error:

$$f(v) = \frac{1}{2}\|Zv - y\|^2$$

- Notice that in both cases the target is still 'y'.

Degree of Polynomial and Fundamental Trade-Off

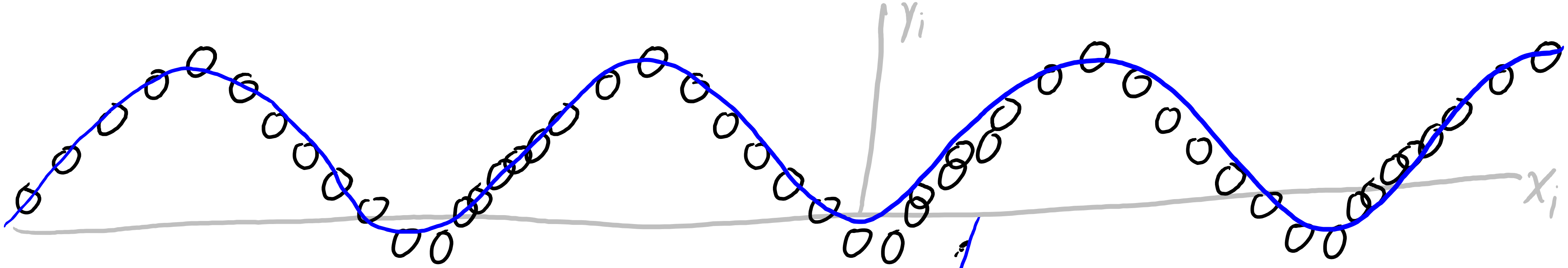
- As the polynomial degree increases, the training error goes down.



- But generalization gap goes up: we start overfitting with large ‘p’.
- Usual approach to selecting degree: validation or cross-validation.

Beyond Polynomial Transformations

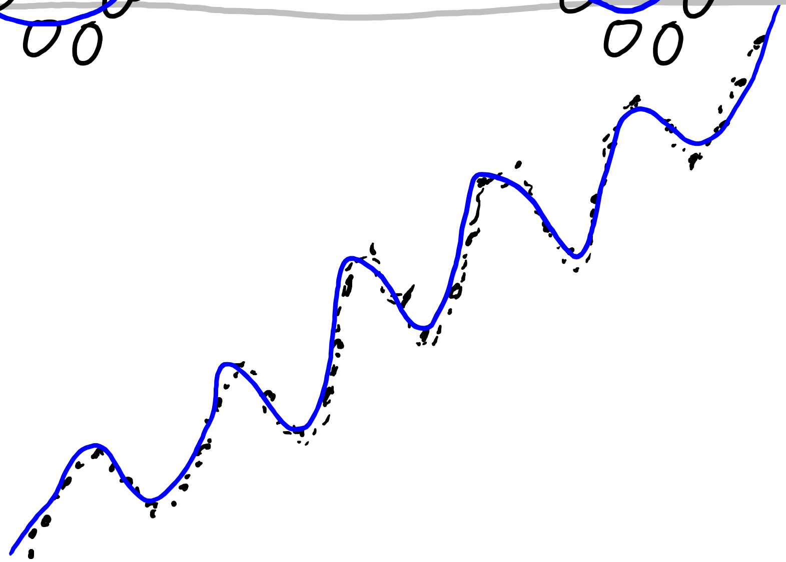
- Polynomials are not the only possible transformation:
  - Exponentials, logarithms, trigonometric functions, and so on.
  - The right non-linear transform will vastly improve performance.
    - Later we will see "deep learning" where you try to learn a transformation.



For periodic data we might use

$$Z = \begin{bmatrix} \sin(x_1) \\ \sin(x_2) \\ \vdots \\ \sin(x_n) \end{bmatrix}$$

$$\hat{y}_i = v^T z_i$$
$$= w_1 \sin(x_i)$$



xkcd

You can have different types of bases

$$Z = \begin{bmatrix} x_1 & \sin(6x_1) \\ x_2 & \sin(6x_2) \\ \vdots & \vdots \\ x_n & \sin(6x_n) \end{bmatrix}$$

$$\begin{array}{cc} \text{linear} & \text{periodic} \end{array}$$

Summary

- Matrix notation for expressing least squares problem.
- Normal equations: solution of least squares as a linear system.
  - Solve $(X^TX)w = (X^Ty)$.
- Solution might not be unique because of collinearity.
  - But any solution is optimal because of "convexity".
- Tree/probabilistic/non-parametric/ensemble regression methods.
- Non-linear transforms:
  - Allow us to model non-linear relationships with linear models.

- Next time: how to do least squares with a million features.

Linear Least Squares: Expansion Step

Want 'w' that minimizes

$$f(w) = \frac{1}{2}\sum_{i=1}^{n}(w^Tx_i - y_i)^2 = \underbrace{\frac{1}{2}\|Xw - y\|_2^2}_{\text{Let's expand then compute gradient.}} = \frac{1}{2}(Xw-y)^T(Xw-y)$$

$$= \frac{1}{2}((Xw)^T - y^T)(Xw - y)$$

$$= \frac{1}{2}(w^TX^T - y^T)(Xw - y)$$

$$= \frac{1}{2}(w^TX^T(Xw-y) - y^T(Xw-y))$$

$$= \frac{1}{2}(w^TX^TXw - w^TX^Ty - y^TXw + y^Ty)$$

$$= \underbrace{\frac{1}{2}w^TX^TXw - w^TX^Ty + \frac{1}{2}y^Ty}_{\text{Sanity check: all of these are scalars.}}$$

Rule:

$\|a\|^2 = a^Ta$

$(A+B^T) = (A^T+B^T)$

$(AB)^T = B^TA^T$

$(A+B)C = AC+BC$

$A(B+C) = AB+BC$

$\underbrace{a^T}_{\text{vector}}A\,b = \underbrace{b^T}_{\text{vector}}A^Ta$

question · stop following · 81 views

Norms Norms Norms: Getting from Sums to Norms

I was going over the solutions for A3 and I am still a bit confused on how to get from a sum to a norm in some situations. I know the basic ones that give me $||Xw-y||^2$ and stuff, but when other things are thrown in the mix I get a bit confused. For example, $\sum_{i=1}^n v_i(w^Tx_i - y_i)^2$ gives $||V^{1/2}(Xw-y)||^2$. From my understanding v is a vector and vi is the number at position I in that vector. How does the summation of these indices result in the diagonal matrix V and not just the vector v?

Furthermore, when we have a summation like $\sum_{j=1}^d \lambda_j|w_j|$, it is simplified to $||\Lambda w||_1$. How does the lambda end up inside the L1 norm? I thought that a summation could be simplified to a L1 norm if its terms are wrapped around the absolute value symbol. In this case the lambda is not, so how is it able to appear inside the norm like that?

hw3 midterm_exam

the instructors' answer, *where instructors collectively construct a single answer*

I know that this notation seems intimidating if this is the first time you see it. Fortunately, there are really only a few "rules" you need to figure out, and you'll find that these are use all over the place.

For those particular questions you'll want to memorize the way that the three common norms appear: $\sum_{i=1}^n |r_i| = ||r||_1, \sum_{i=1}^n r_i^2 = ||r||^2, \max_{i\in\{1,2,\dots,n\}}\{|r_i|\} = ||r||_\infty$. So when you see max, sum of non-negative values, or sum of squared values you should think of these norms.

Next, notice what multiplying by a diagonal matrix does: if you multiply a vector w (for example) by a diagonal matrix then you multiply each element w_i by the corresponding diagonal element. If you multiply matrix X (for example) by a diagonal matrix then you multiply each row of X by the corresponding diagonal element.

The $V^{1/2}$ shows up because we're multiplying the square.

The other really useful ones to know are $\sum_{i=1}^n v_i r_i = v^T r$ if the elements aren't necessarily non-negative, $\sum_{i=1}^n \sum_{j=1}^n a_{ij}x_i x_j = x^T A x$, and $\sum_{i=1}^n x_i r_i = X^T r$.

(All of the above follow from definitions, but it takes some practice to recognize these common forms. That's why we made you get some practice on the assignments, and why we covered this notation before the midterm so that you study it before we start using it a lot. It is incredibly common the ML world.)

edit · good answer | 2 Updated 8 months ago by Mark Schmidt

Bonus Slide: Householder(-ish) Notation

- **Househoulder notation:** set of (fairly-logical) conventions for math.

Use greek letters for scalars: $\alpha = 1$, $\beta = 3.5$, $\gamma = \pi$

Use first/last lowercase letters for vectors: $w = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix}$, $x = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $y = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $a = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$

→ Assumed to be column-vectors.

Use first/last uppercase letters for matrices: X, Y, W, A, B

Indices use i, j, k.

Sizes use m, n, d, p, and k

hopefully meaning of 'k' is obvious from context

Sets use S, T, U, V

Functions use f, g, and h.

When I write x_i I mean "grab row 'i' of X and make a column-vector with its values."

Bonus Slide: Householder(-ish) Notation

- Househoulder notation: set of (fairly-logical) conventions for math:

Our ultimate least squares notation:

$$f(w) = \frac{1}{2}\|Xw - y\|^2$$

But if we agree on notation we can quickly understand:

$$g(x) = \frac{1}{2}\|Ax - b\|^2$$

If we use random notation we get things like:

$$H(\beta) = \frac{1}{2}\|R\beta - P_n\|^2$$

Is this the same model?

When does least squares have a unique solution?

- We said that least squares solution is not unique if we have repeated columns.
- But there are other ways it could be non-unique:
  - One column is a scaled version of another column.
  - One column could be the sum of 2 other columns.
  - One column could be three times one column minus four times another.

- Least squares solution is unique if and only if all columns of X are “linearly independent”.
  - No column can be written as a “linear combination” of the others.
  - Many equivalent conditions (see Strang’s linear algebra book):
    - X has “full column rank”, X^TX is invertible, X^TX has non-zero eigenvalues, $\det(X^TX) > 0$.
  - Note that we cannot have independent columns if $d > n$.

Vector View of Least Squares

- We showed that least squares minimizes:

$$f(w) = \frac{1}{2}\|Xw - y\|^2$$

- The ½ and the squaring don’t change solution, so equivalent to:

$$f(w) = \|Xw - y\|$$

- From this viewpoint, least square minimizes Euclidean distance between vector of labels ‘y’ and vector of predictions Xw.